

Linux Introduction and basic commands

Class no:9

What is Linux?

- **Linux Kernel:** At its core, Linux is a kernel (the central part of an OS that manages hardware and system resources).
- **Operating System Family:** When combined with system utilities, libraries, and applications, it becomes a full operating system distribution.
- **Open Source:** Developed initially by Linus Torvalds in 1991, Linux is free and open source, meaning anyone can modify and distribute it.
- **Unix-like:** It is modeled after Unix, offering stability, multitasking, and multi-user capabilities.

Is Linux an Operating System?

- Strictly speaking, Linux itself is a kernel.
- However, in everyday usage, people call “Linux” an operating system because distributions package the kernel with essential software to make a complete OS.
- Example: Ubuntu = Linux kernel + GNU utilities + desktop environment + package manager.

Popular Linux Distributions

Distribution	Based On	Key Features	Best For
Ubuntu	Debian	User-friendly, large community, strong support	Beginners, desktops, servers
Fedora	Independent	Cutting-edge, backed by Red Hat	Developers, testing new features
Debian	Independent	Stability, huge software repository	Servers, advanced users
Arch Linux	Independent	Rolling release, highly customizable	Power users
Linux Mint	Ubuntu/Debian	Easy to use, Windows-like interface	Beginners switching from Windows

Why Linux is Popular ?

- **Free & Open Source:** No licensing costs.
- **Secure & Stable:** Less prone to malware compared to Windows.
- **Flexible:** Runs on desktops, servers, mobile devices, and even supercomputers.
- **Community Support:** Thousands of contributors worldwide.
- **Performance:** Lightweight distributions can run on older hardware.

Key Notes

- Linux is not a single OS but a family of operating systems built on the Linux kernel.
- Choosing a distribution depends on your needs: Ubuntu/Mint for beginners, Fedora/Debian for stability, Arch for customization.
- Linux powers everything from Android phones to supercomputers.

Linux File System Directories:

The Linux file system, starting from the **root directory** (/), organizes everything on a Linux machine (0:21). This structure ensures consistency across different Linux distributions by following the **Filesystem Hierarchy Standard (FHS)**.

Here's an overview of the key directories:

- **Root and User-Related Directories:**
 - / (Root Directory): The top-level directory from which all other files and directories descend. Unlike Windows, Linux has a single unified file system where everything connects back to root.
 - /root: The personal home folder of the root user, who has full administrative access.
 - /home: The personal space for regular users, where each user has their own subdirectory for documents, downloads, and personal settings.
- **System Binary & Library Directories:**
 - /bin: Contains essential command-line tools needed for the system to function properly, especially during boot or recovery mode. Examples include ls, cp, mv, and cat.
 - /sbin: Holds specialized system utilities mostly used by the system administrator or root user for managing and maintaining the system.
 - /lib: Stores shared libraries that support essential programs and commands found in /bin and /sbin. It also contains kernel modules.
 - /usr: Stands for Unix system resources and contains most of the system's user-space applications, libraries, and documentation. It mirrors several directories from the root level, like /usr/bin, /usr/lib, and /usr/sbin.
- **Boot, Kernel & Device Directories:**
 - /boot: Contains all the files your system needs to boot up, including the Linux kernel, initial RAM disk (init RD), and GRUB boot loader files.
 - /dev: Contains device files that act as interfaces to your hardware. In Linux, everything is treated like a file, even devices.
 - /proc: A virtual file system that provides a dynamic, in-memory view of the system, acting as a live interface to the Linux kernel and running processes.
 - /sys: Another virtual directory part of the sysfs pseudo file system, acting as a dynamic interface between the Linux kernel and your hardware.
- **Configuration & Temporary Directories:**
 - /etc: The central location for system-wide configuration files, covering everything from network settings and user accounts to startup scripts.

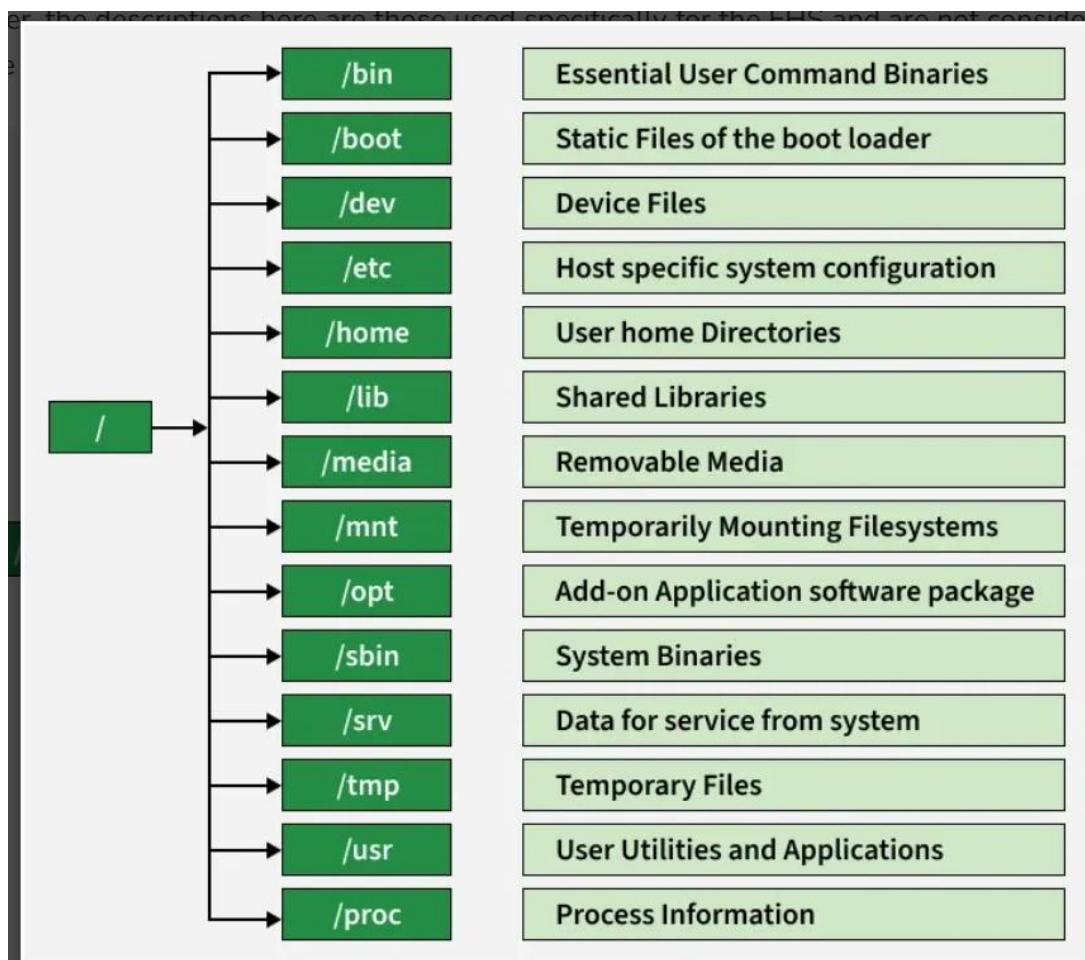
- `/run`: An in-memory directory that holds temporary runtime data used by the system and applications right after boot.
- `/tmp`: Designed for short-term use, this is where temporary files created by applications or users go. Files here are typically deleted automatically on reboot or after a certain period.
- `/var`: Used for storing files that change frequently during system operation, such as logs, caches, and spool files.

- **Mount Point & System Information Directories:**

- `/media`: Where modern Linux systems automatically mount removable media like USB drives, CDs, and DVDs.
- `/mnt`: Traditionally where system administrators would manually mount external storage like USB drives, partitions, or remote file systems.

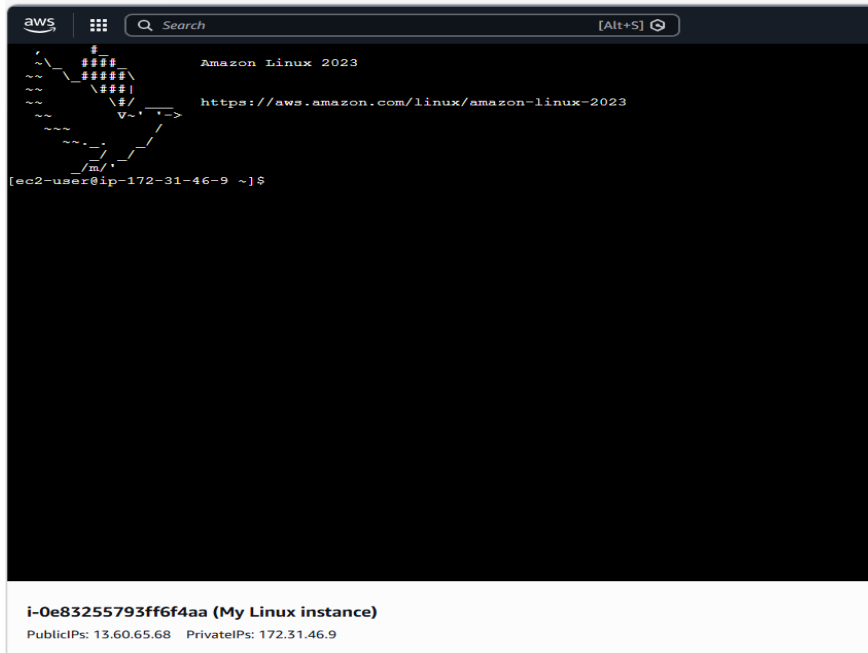
- **Optional Software & Service Directories:**

- `/opt`: Where optional or third-party software is installed, often kept separate from the system's main directories
- `/srv`: Where data for services provided by the system (like web servers or FTP servers) is stored.



Hands on: Basic commands of Linux and Linux file systems:

Step no:1 Create a Linux instance on EC2. Then connect it.



Step no:2 Now we can use some basic commands.

- `pwd` – print working directory.
- `ls` – to list the files in the folder
- `cd` – to change directory of the folder
- `cd ..` – to go one step back

```
[ec2-user@ip-172-31-46-9 home]$ pwd
/home
[ec2-user@ip-172-31-46-9 home]$ ls
ec2-user
[ec2-user@ip-172-31-46-9 home]$ cd ..
[ec2-user@ip-172-31-46-9 /]$ pwd
/
[ec2-user@ip-172-31-46-9 /]$
```

- `cd /` – to go to the root folder
- `cd /folder name` – absolute path to the folder
- `cd folder name` – relative path

```
[ec2-user@ip-172-31-46-9 /]$ cd /home
[ec2-user@ip-172-31-46-9 home]$ pwd
/home
[ec2-user@ip-172-31-46-9 home]$ cd /
[ec2-user@ip-172-31-46-9 /]$ pwd
/
[ec2-user@ip-172-31-46-9 /]$ ls
bin boot dev etc home lib lib64 local media mnt opt proc root run sbin srv sys tmp usr var
[ec2-user@ip-172-31-46-9 /]$ cd local
[ec2-user@ip-172-31-46-9 local]$ pwd
/local
[ec2-user@ip-172-31-46-9 local]$ cd /
[ec2-user@ip-172-31-46-9 /]$ cd /sbin
[ec2-user@ip-172-31-46-9 sbin]$ pwd
/sbin
[ec2-user@ip-172-31-46-9 sbin]$
```

- whoami – to know the username
- sudo su – switch root user (\$ - home user , # - root user ; / - root directory, ~ - home directory)
- su username – to switch back from root user to home user.

```
[ec2-user@ip-172-31-46-9 sbin]$ whoami
ec2-user
[ec2-user@ip-172-31-46-9 sbin]$ sudo su
[root@ip-172-31-46-9 sbin]# whoami
root
[root@ip-172-31-46-9 sbin]# sudo su
[root@ip-172-31-46-9 sbin]# exit
exit
[root@ip-172-31-46-9 sbin]# su ec2-user
[ec2-user@ip-172-31-46-9 sbin]$ whoami
ec2-user
[ec2-user@ip-172-31-46-9 sbin]$
```

```
[ec2-user@ip-172-31-46-9 ~]$ cd ~/
[ec2-user@ip-172-31-46-9 ~]$ pwd
/home/ec2-user
```

- cat – to see the file contents without opening it.
- vim – to create and open a file.

```
bin boot dev etc home lib lib64 local media mnt opt
[ec2-user@ip-172-31-46-9 /]$ cd bin
[ec2-user@ip-172-31-46-9 bin]$ ls
'[' comm gpg-error
ac command gpg2
acpi_listen coredumpctl gpgme-json
addr2line cp gpgv2
alias cpio gpic
amazon-ssm-agent cpupower gprof
```

```
[ec2-user@ip-172-31-46-9 bin]$ cat alias
#!/usr/bin/sh
builtin alias "$@"
[ec2-user@ip-172-31-46-9 bin]$ cat comm
ELF> *
```

```
[ec2-user@ip-172-31-46-9 ~]$ cd /
[ec2-user@ip-172-31-46-9 /]$ pwd
/
[ec2-user@ip-172-31-46-9 /]$ ls
bin boot dev etc home lib lib64 local
[ec2-user@ip-172-31-46-9 /]$ cd mnt
[ec2-user@ip-172-31-46-9 mnt]$ vim mnt
[ec2-user@ip-172-31-46-9 mnt]$
```